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# REINFORCEMENT LEARNING

## Semi-Markov Decision Processes

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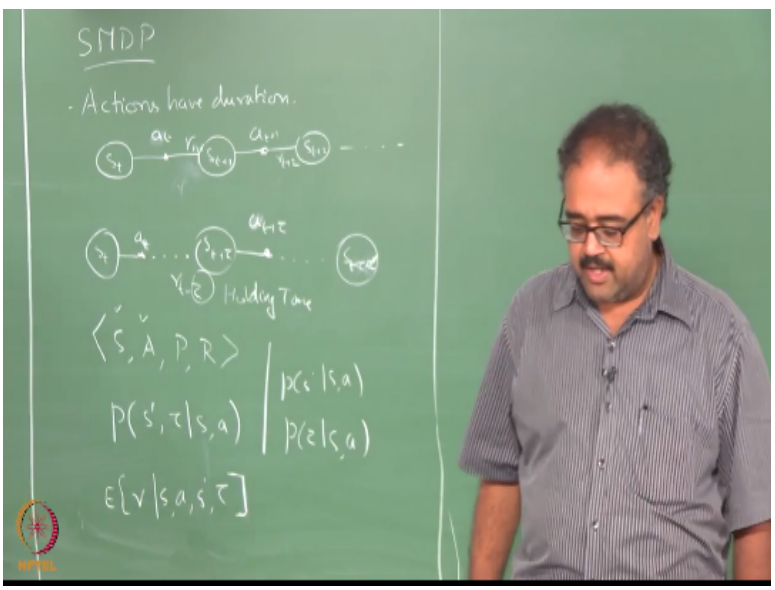


So I said recursive optimality is good if you have a mechanism by which you can treat the solutions that you have as action. So that your horizon your learning horizon becomes smaller right I do not really have to look for many, many, many actions to come before I make a, before I can learn something new right.

Otherwise I said it will not be a very useful thing right, so one of the foundational structures that we use for doing that is called a Semi-Markov Decision process. So what is a Semi-Markov decision process or a Semi-Markov decision process, no that is just non Markov, nth order markov, it depends on two previous states it is second order Markov.

So in Semi-Markov decision processes right, so in a Markov process what do you have so I have a state $S_t$, I take an action as $a_t$, then I go to $S_{t+1}$, I get $R_{t+1}$ right. So and this happens again and again, so next I take an action right it keeps going .. here I have to see $R_{t+2}$, so you guys can see my font from there right okay.

So in Semi-Markov decision process what happens is I am in $S_t$, I take $a_t$, then some time elapses I would find myself in the next state at time $t+\tau$ where $\tau$ is some kind of a random variable that is sample based on $S_t$ and $a_t$ right I also get some reward $r_{t+}\tau$. Now I take another action, it is not $t+2\tau$ okay so it is $t+ \tau+ \tau'$ we are bothered about the time right.

So because we use $\gamma^t$ sorry, so I will use $\gamma^t r_{t+1}$, right $\gamma^1$ basically right I will use $\gamma \times r_{t+1}$ or no, no $\gamma \times r_{t+2}$ , $\gamma^2 r_{t+3}$ and so on so forth. Now what I will do is I will use $\gamma^\tau \times r_{t+\tau} + \gamma^{\tau+\tau'} \times r_{t+\tau+\tau'}$ and so on so forth.

And the next time I do the same actions right the the time taken might be different. So now I am not only interested in what the next state I am going to land up in is how long will it take me to get to the next state because that is how much I will discount the reward by, how do you get to sub problem.

No maybe that is the best way to solve the subproblem it takes you five time units to solve one sub problem right. So I am not even talking about any sub problems here by the way you are getting ahead of yourself okay, this is just a Semi-Markov decision process there are no sub problems here, there are only states and actions okay, there are no subproblems here I am just saying actions take different amounts of time to complete.

So optimality now would depend not only on where the action takes me but how long it takes to, how long it takes to get there right. What is it, explain the $\tau$, $\tau$ is just the time taken for the action to complete it is a random variable, so every time I take an action it may take a differing amount of times to complete.

Dependent on $\tau$, in reality it would in many cases it would, but we kind of wave our hands here and decouple the $\tau$ and S' distributions right. So typically we end up decoupling the $\tau$ and S' distribution so it makes it little easier, but typically it should be a joint distribution right. So historically SMDP models used to do that right, but then Tom Dietterich introduced a variant of the SMDP model into the literature.

Specifically meant for modeling hierarchies right where he actually defines the distribution as joint distribution over $\tau$ and S' right. So historically, so people used to talk about this as holding time, I keep saying historically people who use SMDPs in OR and other things will kill me because that is how they use it even today right.

So it is called holding time right, so holding time essentially the idea is that at time t, I decide a is the action I am going to take right, but I do not apply the action a till $t+\tau$. It could be a variety of reasons I mean I could actually be sensing the market at some point of time and by the time I place my order it might be 15 seconds later or 20 seconds later or 30 seconds later there could be variety of things that come in.

There could be many reasons why you want to do that, I mean you could, there should be gap between the sensory input coming to you and you taking an action right. So it is Semi-Markov in the sense that for the duration of the action I do not know how the system is evolving right. So to determine when the next state is going to occur I really need to know how much time has elapsed since the last state was seen right.

So some amount of history dependence is there, so that is why it is called Semi-Markov right. So I know that $\tau$ seconds have to elapse before the next state happens right, but how much of the tau seconds is left right. So that is why it becomes Semi-Markov not completely non Markov because once this is happened the probability of $S_{t+\tau}$ depends only on $S_t$ and $a_t$ right.

But during this transition when this will actually show up right depends on how much time has elapsed okay. So it is only for this brief periods you have the time dependency okay.

That is just semantics right, so some people refer to it as holding time, holding time is when you actually, when do you apply the action right other one is you can think of it as transition time. So I apply the action immediately how long does it take for the next state to actually happen that is like a transition time right, both mean the same thing.

So for RL people its transition time makes more sense right, but in the other the holding time the explanation was given only because of the semantics of things right. So what really happens if I already applied the action what happens to the system after the action is applied. So instead of asking that question you say okay I will stay in this state for tau seconds and then I will apply the action immediately I will get the next state.

So there you kind of side step the question of what happens after you apply the action and before the next state turns up. But we do not want to sidestep the question, I apply the action go to room 1 and I want to know what happens after that right. So I will pay attention to this, that is why the transition time definition is more acceptable for RL folks right.

So we have some more time so the SMDP is defined as the $\langle S, A, P, R \rangle$ see the thing is these are explanations these are models that you have right, so this is at the end of it mathematically what is happening is there is a state that occurs at time t, there is a state that appears at time $t+\tau$ okay that is it.

So what happens in between people come up with different ways of interpreting that see one way that says that okay I actually apply the action at time t itself and it takes a system some time to evolve and settle down into $S_{t+\tau}$. In what way, …no no no generally we assume that the state stays for time tau that is why it is the holding time so you come to a state you hold the state for time $\tau$

seconds and then I apply the action I go to the new state. So as soon as I apply action I, you go to the new state.

The other one is, but in some cases we will have both kind of delays in. It does not matter see the point about SMDP is I really do not tell you what happens in those τ seconds right and you can interpret it different ways but mathematically you end up with the same thing, see the quantities I am using for modeling as I will write it down now you will see what I am going to say that eventually end up with the same thing right.

In fact the holding time interpretation says the following, I see $S_t$, I pick a t immediately, but I wait for τ seconds before I apply it right. So the decision is made before the waiting time starts, before the holding time starts. So that the holding time is still a function of the decision okay. So I have a $\langle S, A, P, R\rangle$ so you know S we know A so we know all of this from the standard thing.

So P is defined as the following, P I get, I define $P(S', \tau|S, a)$. So earlier our probability distribution was $P(S'|S, a)$ but now $P(S', \tau|S, a)$. And the reward becomes $E[r|s, a, s', \tau]$ so why does the $\tau$ figure in, I mean because it will actually end up getting some kind of discounting on it.

So it could be accumulating some rewards along the way, it could be just the waiting time so for every second you wait, you accrue reward at a rate of -1 per second or something right so you could give some kind of a rate of reward right. So it could be some R, -R is the rate of reward and then if I am waiting for $\tau$ seconds and basically I do an integral over that to get you the total sum of the reward right the rate is –R right.

So I could get some things like that right so the the reward that I get could depend on the the transition time or the holding time and so on so forth. So earlier people used to write this as two different functions, $p(s'|s, a)$ and $p(\tau|s, a)$ so they, right notice that this is not a perfect decomposition of that right I am assuming τ is independent of S' I actually had used the product rule to decompose it $\tau$ should have been dependent on S' or S' should have been dependent on $\tau$, one way or the other right.

So but now I am just saying this, so this is the classical way let me not put it the earlier way, it is the classical way of writing this SMDP transition dynamics as into two components one on the $\tau$, one on the S' yeah, but this is more meaningful for the RL domain but it is more meaningful for RL domain because depends on what is the problem that where I am going to land up in right the time that I am going to take.

For example, if I am going to go out through A or go out through B right the time taken will be different for executing it start from the same state if I am choosing to go through A it will have some time if I am choosing to go through B it will have a different time. So depending on which S' I am going to go for the $\tau$ will be different right.

So those things should matter right, so I cannot do a separate distribution like this I have to do a joint distribution over S' and $\tau$ okay. Any questions on SMDPs all of that is clear right you can write SMDP value functions, you can write SMDP bellman equations I hope all of you can write that, can you write SMDP bellman equations, well depends if you can write SMDP value functions first right.

Write the value function and then write the recursive version of the value function right, and then then convert that into a bellman equation how would I decide on the what $\gamma$, $\gamma$ is well that is the same problem with all discounted reward formulations right, $\gamma$ has to come from the domain expert yeah the expectation has to be over the $\tau$ also.

So it becomes.. that is not very hard it is a little tricky to write this thing, but it is not too hard right so I am leaving it as an exercise, I am going to leave all of you to go and look up bellman optimality equation or rather the bellman equation for SMDP can you put that recent advances in hierarchical RL paper Barto and Mahadevan right.

So there are lot of hierarchical RL frameworks that have been proposed in the literature and surprisingly you would get more information about these frameworks by reading a survey paper that was written a little later after all this hierarchy all the architectures are proposed than reading the original papers themselves right.

So there are two reasons for it, so kind of the authors of the survey paper had a little bit more perspective so they could see all the things together in a unifying framework and they could present it in a better fashion. And of course, I am very biased and therefore the one of the authors of the survey paper is a fantastic writer and therefore he wrote a better paper than the original writers themselves right.

So my advisor so of course I am biased so read that paper so recent advances in hierarchical RL and so he describes SMDP and SMDP value functions and everything very, very beautifully in the paper right. And then I am going to talk about a few other things that we can do first thing I want to talk about is something called SMDP Q learning right.

So what do you think is SMDP Q learning …

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha[\bar{r}_{t+\tau} + \gamma^{\tau} max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)]$$

is it +/-1 do we need a -1 there yeah, in the $\gamma$ you are right $\tau$ time steps if $\tau$ is 1, I do $\gamma$ power.. $\gamma$ right. $\tau$ is 1 I do $\gamma$..okay right so that is basically SMDP Q-learning it looks very simple right it is just that the rewards are discounted.

So the tricky thing here is I have hidden some stuff into this $r_{t+\tau}$ right, so what is $r_{t+\tau}$ right. So it is the expected reward that I am going to get over the tau time steps right it is not the expected reward it is still a sample right I just taken a sample over $\tau$ time steps, so it could be variety of things right so normally in the classical SMDP framework we do not tell you how this $r_{t+\tau}$ comes about.

So you have some mechanism by which you can generate it right it could be some e power r $\tau$ integral 0 to $\tau$, $e^{r_t \int_0^{\tau} d\tau}$ right. So that could be one way of defining this so that is a rate function, so R is the rate function and I accumulate that function for $\tau$ time steps right and that that gives you a reward right.

But what we are more interested in is to actually think of $r_{t+\tau}$ as

$$r_{t+\tau} = r_{t+1} + \gamma r_{t+2} + \cdots + \gamma^{\tau-1} r_{t+\tau}$$

okay, so this $r_{t+\tau}$ is our usual $r_{t+\tau}$ right, the reward the one-step reward that you get τ time steps after you took the action right. So maybe this makes it clearer, so I am not written something that is recursive so maybe there you go. So $\overline{r_{t+\tau}}$ is essentially the return I have accumulated for the τ time steps after I have chosen action $a_t$.

So in some sense if you think about this two world example, so I will start here let us say and then I say do B I mean go to room 2 right, then it is basically the reward I get for till that point right and the value of go to room 2 here will get updated by the value of the state here right. So I will take the Q here and do the $max_a$ right and for the $\overline{r_{t+\tau}}$ I will use all the rewards I get till that point right and I will take both those quantities and then update the Q of go to room 2 here makes sense okay.

So this is called SMDP Q-learning I am sorry, it depends on the reward function, the reward function is the usual -1 per time step reward function then you are minimizing time or if the reward function is also 1 when you reach the goal and γ discount along the way, then also you will be minimizing time except that for this action $r_{t+\tau}$, $\overline{r_{t+\tau}}$ will always be 0 right because you get no reward.

But then this slowly your reward will start propagating back okay. So notice that SMDP Q-learning does not actually look into the structure of a t right, it just assumes that this reward somehow comes when I take action $a_t$ right. So it assumes that $a_t$ has been learnt already so what is $a_t$ here in this example go to room 2, $a_t$ was go to room 2 right.

So it is not a simple action this is what you mean by the temporal abstraction so when you say go to room 2 it actually takes some 10 time steps to go to room 2, if I had said going to room 2 from here it would have just taken two time steps to go to room 2, so it would be different right. And there is some kind of stochasticity in the world right let us say with probability 0.9 my action will succeed right otherwise it will fail.

Then it will take different amount of times even from the same state if I say go to room 2 it will take different amounts of time to complete. So that is basically so we will stop here with SMDP Q-learning. So from next class I will actually look into the different hierarchical architectures that are available and then we will see how we can modify SMDP Q-learning to work with different architectures.

You can choose whatever you want, SMDP Q-learning defines it like this, because it is defined on the discounted return. So eventually at the end of the day you will still want to solve for the discounted return right, because you are trying to optimize the discounted return eventually right. So if you are trying to optimize average return then you can define your reward functions differently.

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